

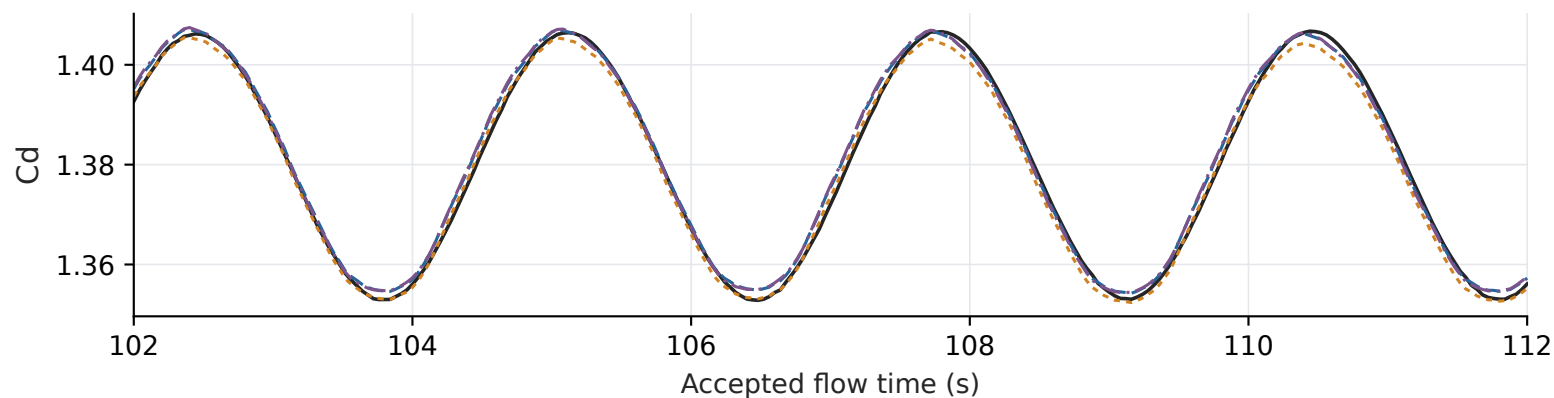
Cylinder boundary-condition controls

Native reference advanced from 100 to 112 s; three isolated FVM controls replay its outer fields

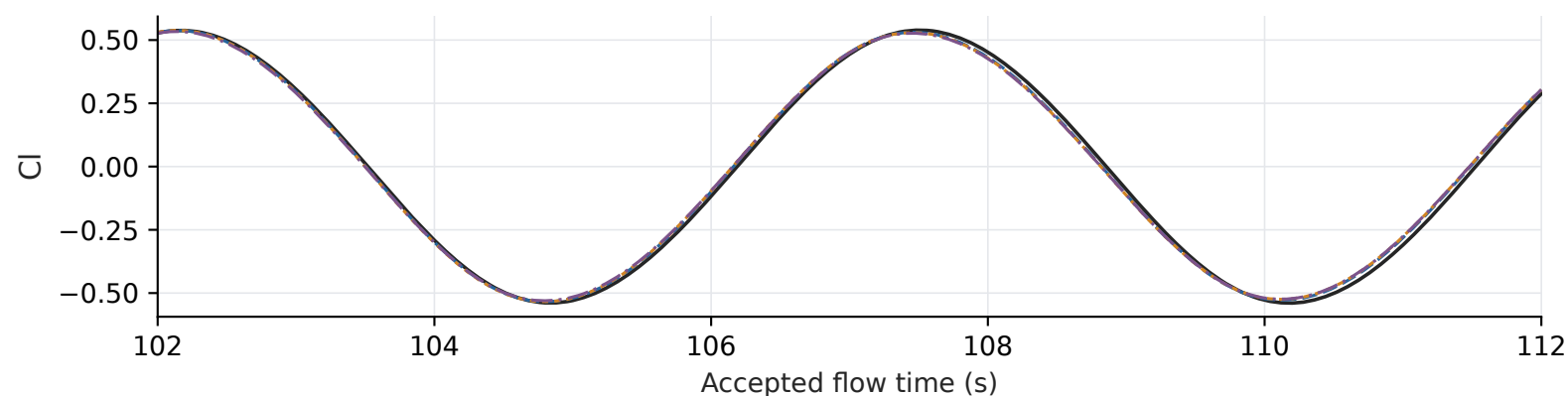
Shown and measured: 102–112 s, after 2 s settling. All retain fixedFluxPressure.

— Reference flow
- - - Full velocity · NumPy · $\Delta t = 0.008$ s
- - - Mixed boundary · NumPy · $\Delta t = 0.008$ s
- - - Mixed boundary · Numba · $\Delta t_{\text{ex}} = 0.04$ s

(a) Cd: reference-driven controls



(b) Cl: reference-driven controls



(c) Forced controls · 102–112 s · complete-cycle medians · 3 Cd cycles / 1 Cl cycle per run

Case	Cd raw	Cd drift-corrected	Cl raw	Cl drift-corrected
Reference flow	0.053701	0.053699	1.078737	1.078404
Mixed · NumPy	0.052404	0.052294	1.064751	1.061576
Full velocity · NumPy	0.052360	0.052309	1.063089	1.058513
Mixed · Numba + subcycling	0.052400	0.052221	1.057401	1.054364

(d) Separate production comparison · 30–44 s · 4 Cd cycles / 2 Cl cycles per run

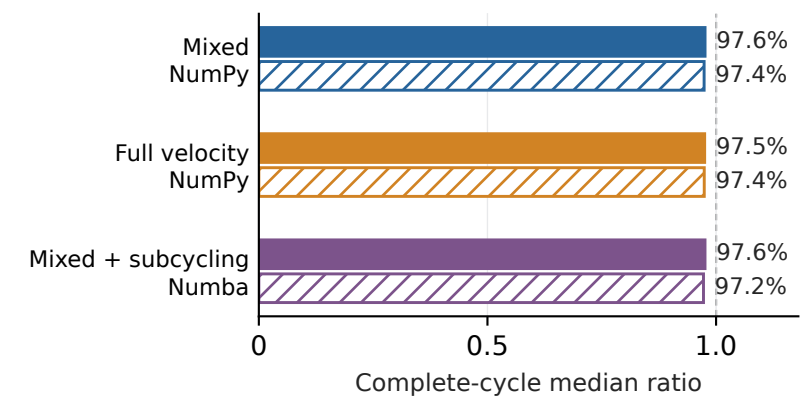
Case	Cd raw	Cd drift-corrected	Cl raw	Cl drift-corrected
Reference flow	0.052607	0.051886	1.054510	1.048950
Actual coupled FVM–VPM	0.035681	0.034720	0.864176	0.857276
Coupled / reference	67.8%	66.9%	82.0%	81.7%

Raw = actual min–max within a complete trough–bracketed cycle; drift-corrected = peak minus interpolated trough baseline.

Bars use control/reference median ratios. Hatched bars are drift-corrected; solid bars are raw. No confidence intervals inferred.

These externally forced controls demonstrate boundary-model capability; they do not demonstrate self-sustained coupled amplitude recovery.

Cd amplitude / reference



Cl amplitude / reference

